Computational Fluid Dynamics
Associate Professorship of Hydromechanics
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Assignment 05

Group 8

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1 Introduction

This assignment addresses the simulation of two flow cases, a shear and channel flow case, which are simulated using 2D numerical approximations to the Navier-Stokes equations for incompressible flow.

The report is organised into the following sections:

1. An introduction to the governing partial differential equations and numerical solution procedure, including
   a) an overview of the Navier-Stokes equations
   b) an overview of the discretisation,
   c) background to the projection method, and
   d) an overview of the direct and iterative solver methods for the pressure correction and related issues.
2. Analysis of the two flow cases, including
   a) numerical experiments and observations, and
   b) a comparison of the direct and iterative solver.

1.1 Navier-Stokes equations

The Navier-Stokes equations for an incompressible fluid in a two dimensional plane are given by (1) - (3).

$$\frac{\partial u}{\partial t} = \frac{1}{\rho}\frac{\partial p}{\partial x} - u\frac{\partial u}{\partial x} - v\frac{\partial u}{\partial y} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + g_x \tag{1}$$

$$\frac{\partial v}{\partial t} = -\frac{1}{\rho}\frac{\partial p}{\partial y} - u\frac{\partial v}{\partial x} - v\frac{\partial v}{\partial y} + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) \tag{2}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{3}$$

Equations (1) and (2) describe the fluid's motion in the x and y directions via the respective velocities, u and v, the fluid's density, ρ, the kinematic viscosity, ν, the pressure, p, and body forces. Note that there are only body forces in the case of channel flow, which is acceleration due to gravity in the x-direction; this is expressed as g_x.

Equation (3), known as the continuity equation for an incompressible fluid, is a condition at every point in the velocity field that there is zero divergence. When a fluid is incompressible, there is no dependence on the thermodynamic state of the fluid, which is why the conservation of mass can be expressed in this form.

The condition of a divergence-free velocity field is an important condition that can be utilised to compute the pressure of an incompressible fluid. The divergence of (1) leads to a reduction in terms, because the divergence of the transient and the viscous terms must be zero due to (3). The result is the Poisson equation, given by (4).

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = -\frac{1}{\rho}\left(\frac{\partial}{\partial x}\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) + \frac{\partial}{\partial y}\left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right)\right) \tag{4}$$

More generally, the RHS of this equation can be interpreted as the divergence of all but the pressure term, and this quantity can have non-zero values. This means the pressure can be considered as the correction to ensure that the velocity field is divergence free. This important result is the basis for the pressure calculation in the solution method.

1.2 Pressure correction solver

Two different techniques are implemented in this assignment to solve for the pressure correction:

1. a direct solver known as the Gauss-Jordan method, and

2. an iterative solver known as the Gauss-Seidel method.

1.2.1 Gauss-Jordan method

The Gauss-Jordan method directly solves the Poisson equation (6) by matrix inversion. The method performs a modified Gauss-elimination. The dp^{n+1} matrix is reshaped into a vector and the conventional $Ax = b$ linear system of equations is solved, where A is the coefficient matrix and $b = div(\hat{u}^{n+1})$. A must then be inverted and multiplied by the divergence of the velocity, b, to find the dp^{n+1} values.

In the code, the coefficient matrix for the linear system of equations is inverted only once before the first time step. This can be done because the coefficients of the matrix are constant. Consider the discretised Poisson equation given by (9).

$$\frac{\delta t}{\rho}\left(\frac{dp_{i+1,j} - 2dp_{i,j} + dp_{i-1,j}}{\delta x^2} + \frac{dp_{i,j+1} - 2dp_{i,j} + dp_{i,j-1}}{\delta y^2}\right) = \frac{\hat{u}_{i+1,j} - \hat{u}_{i-1,j}}{2\delta x} + \frac{\hat{v}_{i,j+1} - \hat{v}_{i,j-1}}{2\delta y} \tag{9}$$

The left-hand side in (9) contributes terms to the coefficient matrix, so the matrix entries are only related to δx, δy, δt and ρ. These remain constant throughout the solution, i.e. they do not change from one time step to another and there is no adpative time stepping, so the matrix is constant. Regardless of what the right-hand-side value may be, in every time-step the coefficient matrix can be reused. This is an advantage for the following numerical experiments which would not be the case if a variable spatial or temporal discretisation were used.

1.2.2 Gauss-Seidel method

The Gauss-Seidel method is an iterative solver which is used to find an approximation of dp without needing to invert the coefficient matrix. It uses (11) to iteratively solve the linear system of equations over the whole domain for dp^{n+1} until the maximum difference between sequential dp calculations is below a certain predefined tolerance (10) or until a certain number of iterations, k, has been reached.

$$max\left(\left|dp^{k+1}_{i,j} - dp^{k}_{i,j}\right|\right) < \epsilon, \quad \epsilon \triangleq tolerance \tag{10}$$

$$dp_{i,j} = \frac{1}{AP}\left(b_{i,j} - AS \cdot dp_{i,j-1} - AN \cdot dp_{i,j+1} - AW \cdot dp_{i-1,j} - AE \cdot dp_{i+1,j}\right) \tag{11}$$

$$where: \quad AP = -\frac{\delta t}{\rho}\left(\frac{2}{\delta x^2} + \frac{2}{\delta y^2}\right) \quad AN = \frac{\delta t}{\rho}\left(\frac{1}{\delta y^2}\right) \quad AS = \frac{\delta t}{\rho}\left(\frac{1}{\delta y^2}\right)$$
$$AE = \frac{\delta t}{\rho}\left(\frac{1}{\delta x^2}\right) \quad AW = \frac{\delta t}{\rho}\left(\frac{1}{\delta x^2}\right)$$

1.2.3 Checkerboard oscillations

In the discretised Poisson equation, the divergence of the provisional velocity is approximated with a second-order central difference scheme. This method computes the derivatives of the velocity by the velocity values at the neighbouring grid points. This discretisation means there is no dependence of $u_{i,j}$ on the pressure at the same point, $p_{i,j}$, and vice versa. This means there is a decoupling of the odd and even grid points for both the pressure and velocity. If there are fluctuations in the pressure and velocity field with a wavelength of $2\Delta x$, which may come about due to numerical errors, these modes will not be detected by the solver. This leads to a checkerboard pattern of pressure or velocity oscillations in the solution.

The main feature of the numerical method leading to this issue is that the grid is "collocated", referring to the fact that the velocity and pressure values are computed at same grid points or locations. This issue can be resolved by utilising a "staggered" grid, or Rhie-Chow interpolation. Both methods address the problem by achieving a different form of the discretised Poisson equation where there is a coupling of the pressure and velocity at the same points, and this alleviates the checkerboard oscillations.

In this assignment, no attempt has been made to address the problem of using a collocated grid. Therefore, it is expected for some of the numerical experiments that the checkerboard pressure oscillations will be observed.

2 Shear Flow

2.1 Problem description

The first flow scenario examined is a shear flow. A 2D domain was defined on the ranges $x \in [0, 2\pi]$ and $y \in [0, 2\pi]$. Periodic boundary conditions were specified on all boundaries. The velocity field, (u_0, v_0) was initialised as given by (12) and (13)

$$u_0(x, y) = \begin{cases} tanh\left(\frac{15(2y-\pi)}{2\pi}\right) & \text{if } y \leq \pi \\ tanh\left(\frac{15(3\pi-2y)}{2\pi}\right) & \text{if } y \geq \pi \end{cases} \tag{12}$$

$$v_0(x, y) = \frac{1}{20} sin(x) \tag{13}$$

Furthermore, the pressure field was initialised with $p(x, y) = 0$.

Figure 1 shows the initial velocity field. This field has two regions of relatively high $\frac{\partial u}{\partial y}$ at $y = \frac{\pi}{2}$ and $y = \frac{3\pi}{2}$. The shear stress is proportional to the $\frac{\partial u}{\partial y}$, so in these regions of high shear stress will cause rotation to develop in the field.

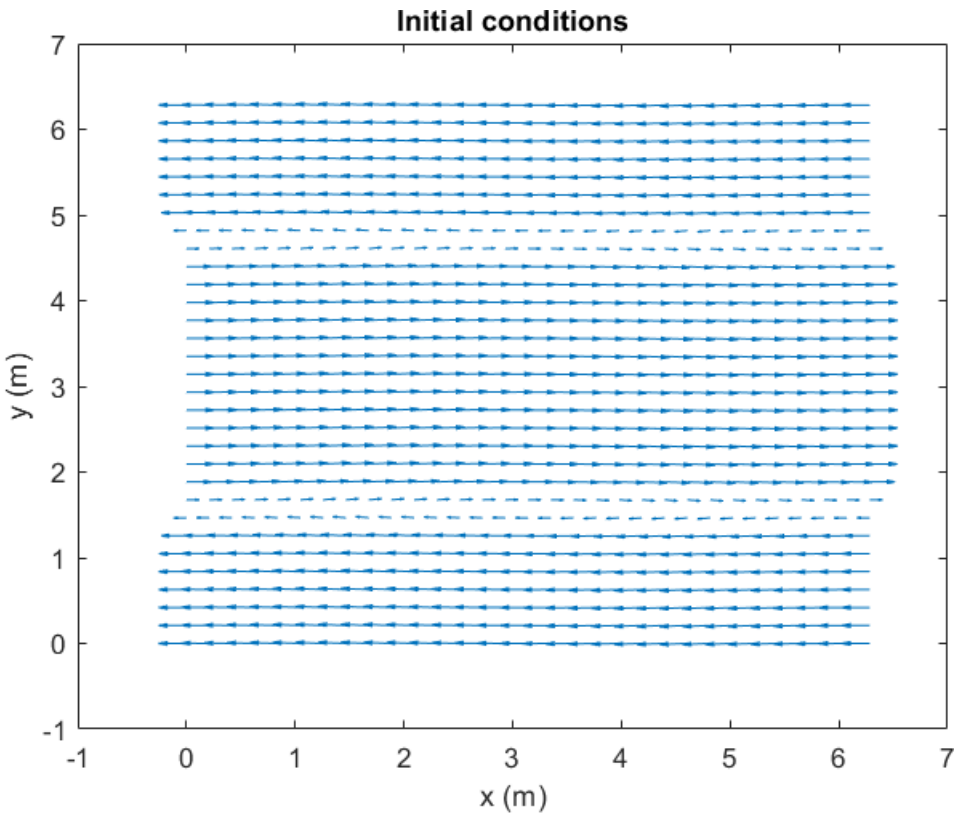

Figure 1: Vector plot of initial velocity field for the shear case.

2.2 Numerical experiments

2.2.1 Direct solver

Firstly, the results obtained using the direct solver are examined. The first numerical experiment was run using $\Delta x = 0.2095m$, $\Delta y = 0.2095m$ (corresponding to a 31 x 31 grid), and $\Delta t = 0.001s$.The kinematic viscosity and density were set as $\nu = 0.01m^2/s$ and $\rho = 1kg/m^3$. The corresponding flow field is shown at different time steps in Figure 2, where the velocity vectors are plotted on top of contours of the 2D velocity magnitude.

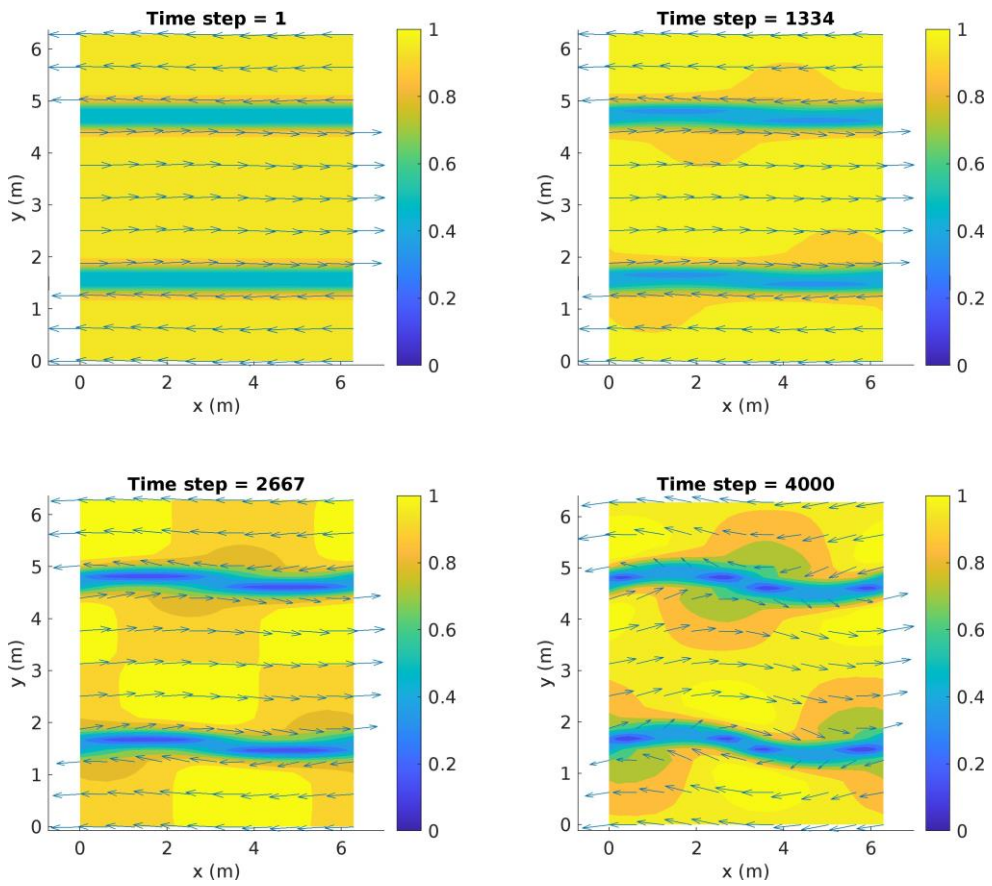

Figure 2: Series of velocity vector plots overlaid on contours of velocity magnitude at different time steps for the reference case.

As expected, the flow starts to develop regions of circulation due to the regions of high shear. The flow is not fully developed after 4000 time steps, as the strong velocity gradients are still present, which will lead to further momentum diffusion.

The pressure field for this flow scenario at the final time step is plotted in Figure 3. Regions of minimum pressure are associated with the centres of the vortices. The pressure field is smooth; no oscillations are present. As mentioned in the introduction, the projection method used for these experiments is susceptible to checkerboard pres-

sure oscillations, but there is no detectable problem in this case because the pressure field is dominated by the presence of the vortices.

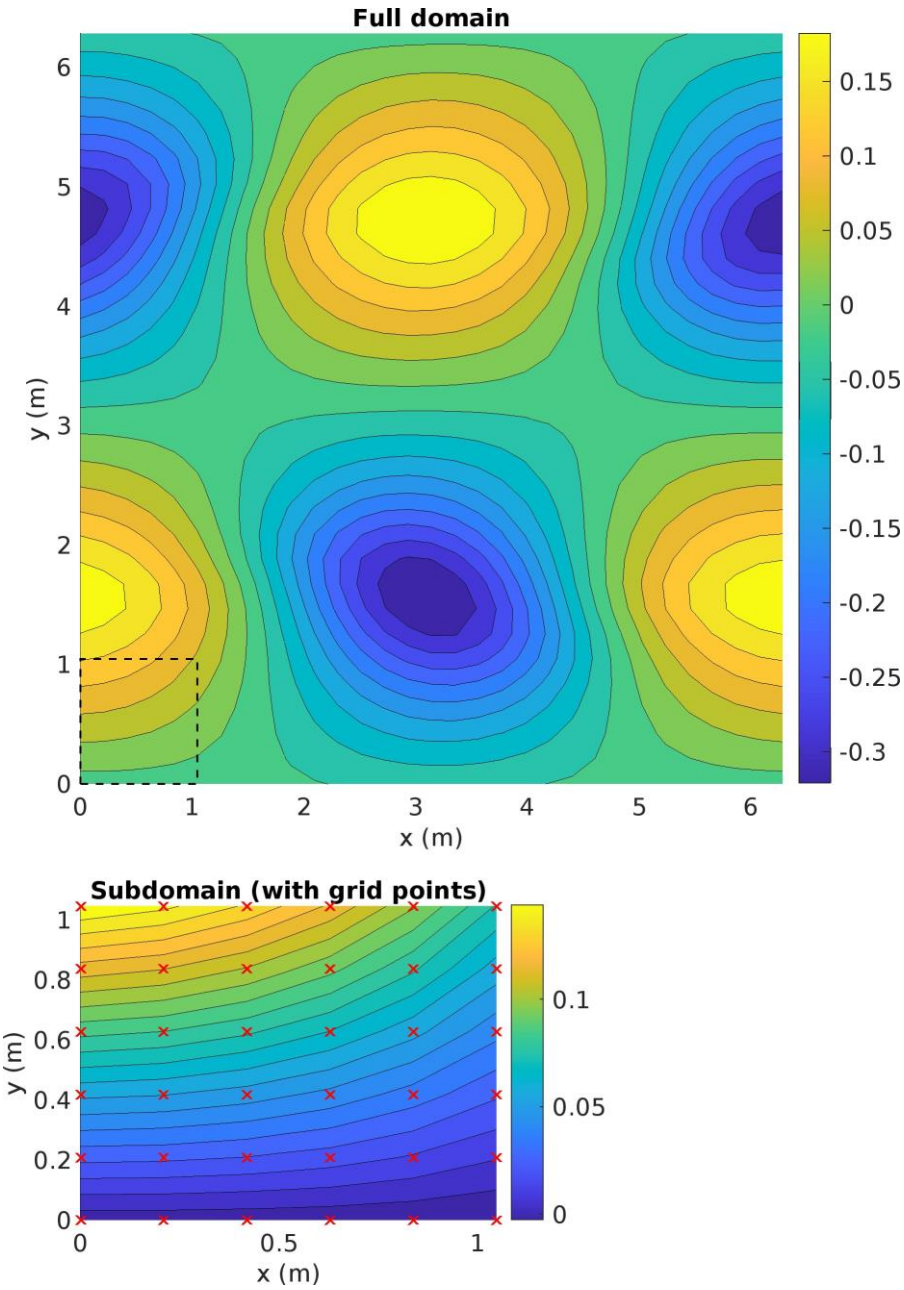

Figure 3: Pressure field for the shear flow, including close-up view at corner of domain

The spatial discretisation was changed to $\Delta x = 0.1257m$, $\Delta y = 0.1257m$ (corresponding to a 51 x 51 grid), and $\Delta t = 0.001s$ was kept the same. From inspection of the solution (plots not included for brevity) in comparison to the first experiment, it was evident that the change in spatial discretisation did not significantly change the solution.

The time discretisation was changed to $\Delta t = 0.01s$. The resulting velocity fields at given time steps are shown as a series of contour plots in Figure 4. These plots show the development over a larger time period, so the circulation is more pronounced. The solution is still bounded/stable and the velocity field is consistent with the expectation that the longer time causes the vortices to become more developed.

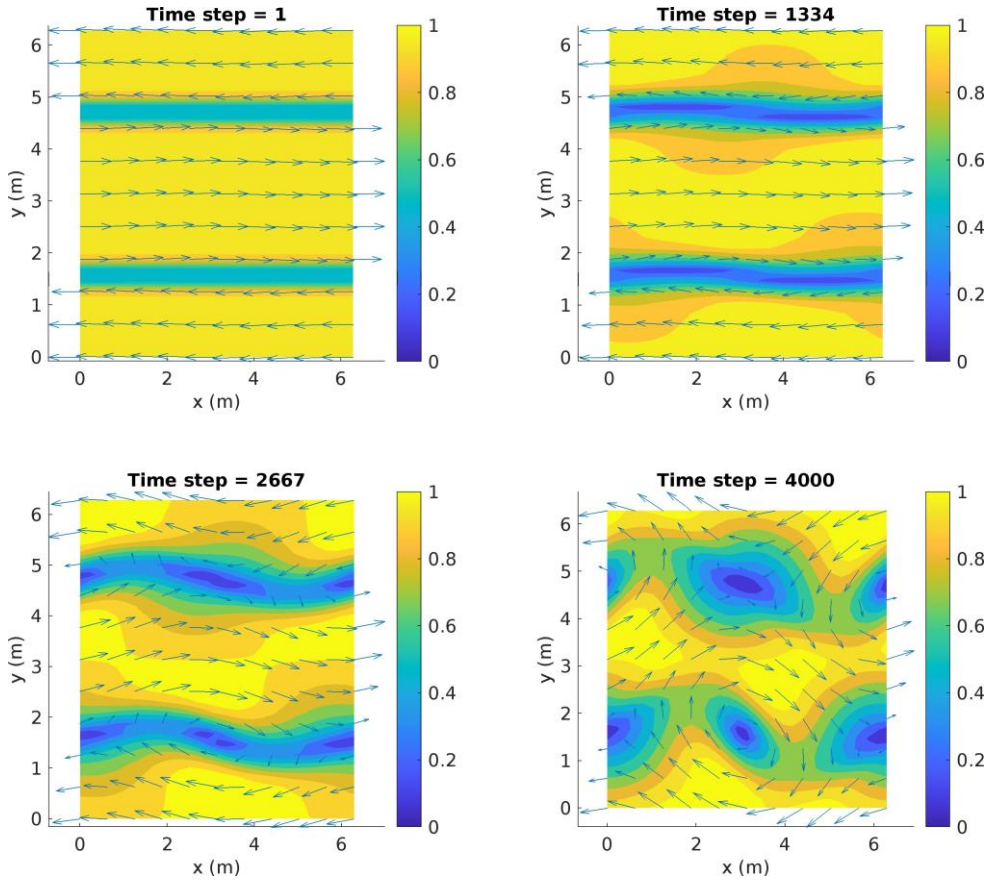

Figure 4: Series of velocity vector plots overlaid on contours of velocity magnitude at different time steps for $\Delta t = 0.002s$

A further experiment was undertaken by varying the density, although no change in solution compared to the reference case was observed. It is noted that the kinematic viscosity was not updated despite the dependency on density.

Reverting back to the discretisation with $\Delta x = 0.2095m$, $\Delta y = 0.2095m$ and $\Delta t = 0.001s$, the effect of the kinematic viscosity was investigated by changing the value to $v = 0.1m^2/s$ and $v = 0.001m^2/s$, respectively. The results are shown in Figure 5. The increased viscosity caused the velocity field to decelerate without leading to the formation of vortices; this can also be interpreted as a reduction in the high velocity gradients with time. With a higher viscosity, the Reynold's number was relatively low for this flow scenario, meaning the flow was laminar and there was no lateral mixing of the layers. Instead, the high shear stress between regions of fluid moving in opposite

directions to each other caused both to lose momentum, and the velocity magnitudes by the end of the simulation, as shown in Figure 5, were significantly reduced.

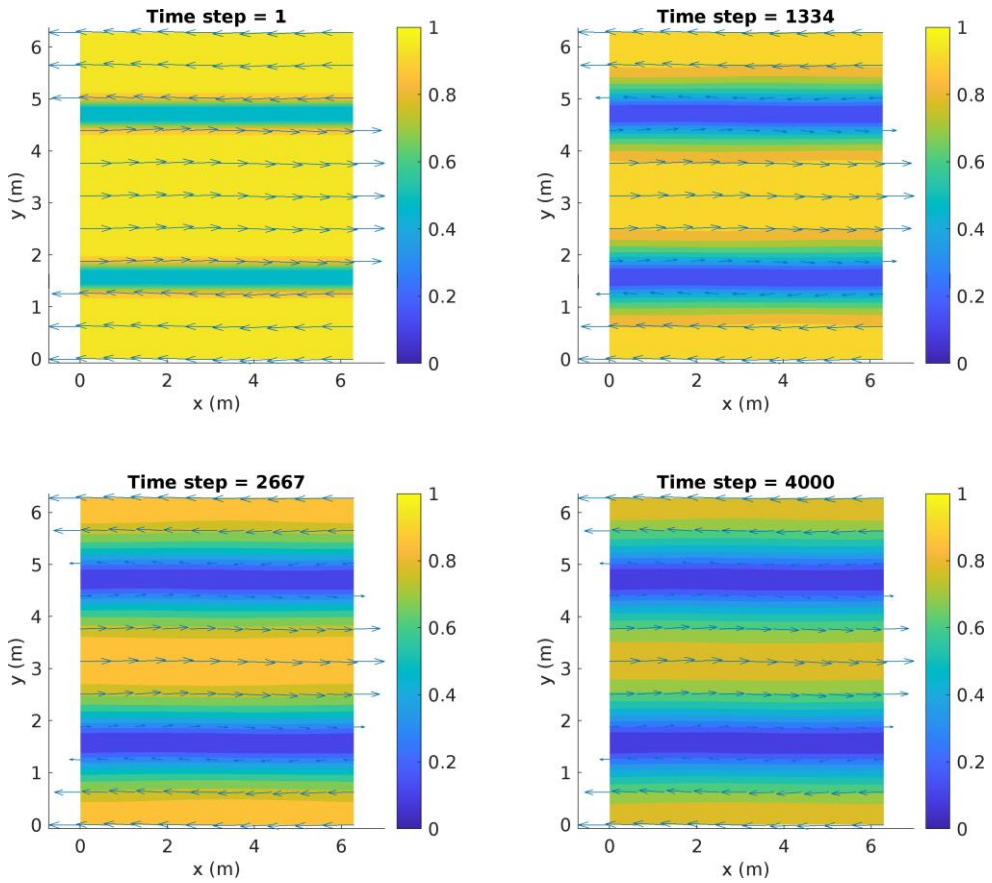

Figure 5: Series of velocity vector plots overlaid on contours of velocity magnitude at different time steps for $v = 0.1 m^s/s$

Reducing the viscosity had the opposite effect, instead causing increased mixing of the horizontal layers but without the reduction in velocities, as displayed in 6. In the series of contour plots, there are visible wiggles in the velocity contour lines, which demonstrates that the velocity is experiencing checkerboard oscillations. The decoupling of odd and even grid points to the pressure lead to grid-dependent fluctuations in the velocity, which is visible as the superposition of small fluctuations within the flow field.

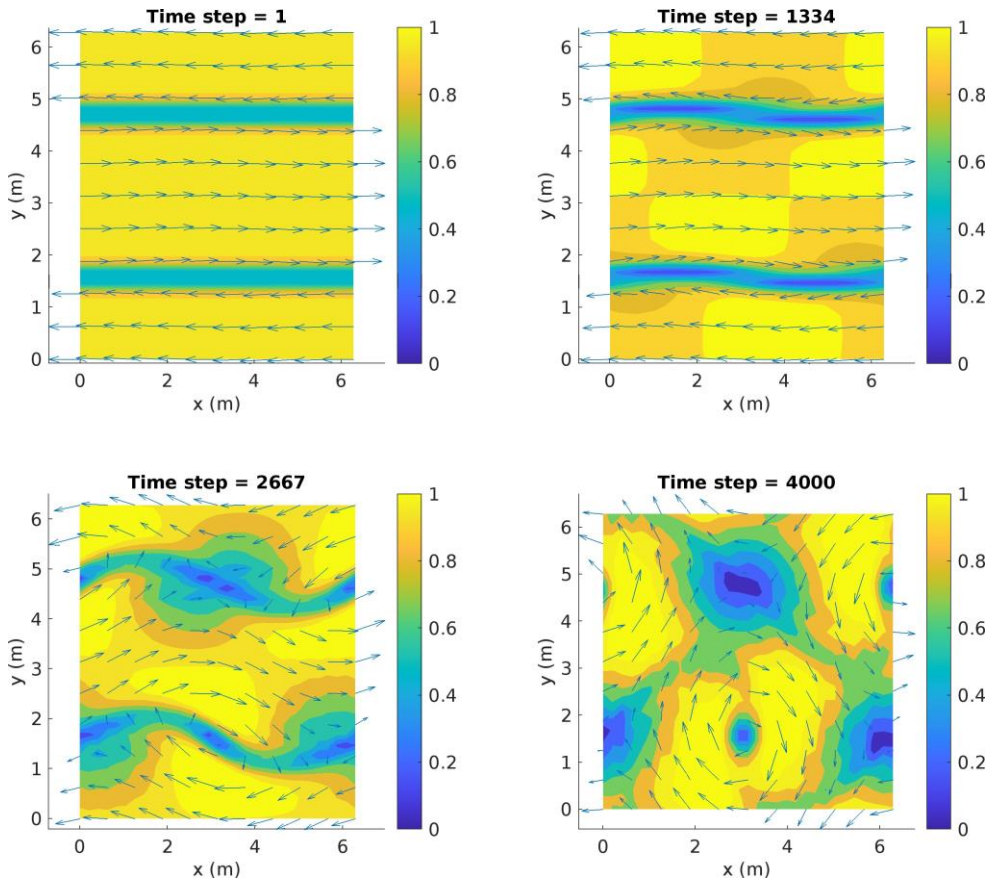

Figure 6: Series of velocity vector plots overlaid on contours of velocity magnitude at different time steps for $v = 0.001 m^s/s$

2.2.2 Direct solver

For each of the flow cases, the iterative solver was run. In all cases, the solution matched that of the direct solver.

The influence of the parameters on the solver performance was analysed by checking the iteration counter for the various numerical experiments. In each case, the tolerance in the stopping criteria of the iterator was set 10^{-4}. The average number of time steps per iteration was determined by averaging the total number of iterations over the total number of time steps (4000). The following points were noted:

1. Increasing the grid resolution (from 31 x 31 to 51 x 51) caused the average number of iterations per time step to increase slightly from 1.07 compared to 1.12.

2. Reducing the time step from $\Delta t = 0.001s$ to $\Delta t = 0.0005s$ caused a slight decrease in the average number of iterations per time step (1.04), but increasing the time step to $\Delta t = 0.002s$ caused a significant increase (2.23).

3. Reducing the kinematic viscosity by a factor of 10 had a larger impact on the

average number of iterations than increasing the viscosity by a factor of 10 (1.47 compared to 1.04).

4. Increasing the density by a factor of 10 had a large impact on the average number of iterations, whereas reducing the density by a factor of 10 caused a slight reduction (5.05 compared to 1.01).

5. Both increasing the density and decreasing the kinematic viscosity by a factor of 10 (i.e. preserving the dynamic viscosity) resulted in an average number of iterations per time step of 7.79.

From these observations, the major influences which caused an increase in the number of iterations required were increasing the time step and increasing the Reynold's number (either by increasing density or reducing viscosity). This is because these factors all cause the dp at each time step be larger. For a larger time step, there is a larger change in the velocity over the time step. For the Reynold's number, the pressure gradient, which is proportional to $1/\rho$, must be larger to enforce the same correction on the divergence of the velocity field. In cases where dp larger, the solver requires more iterations for convergence to the acceptable tolerance.

The number of iterations per time step for channel flow is shown in Figure 7. The results are plotted for three different tolerance values.

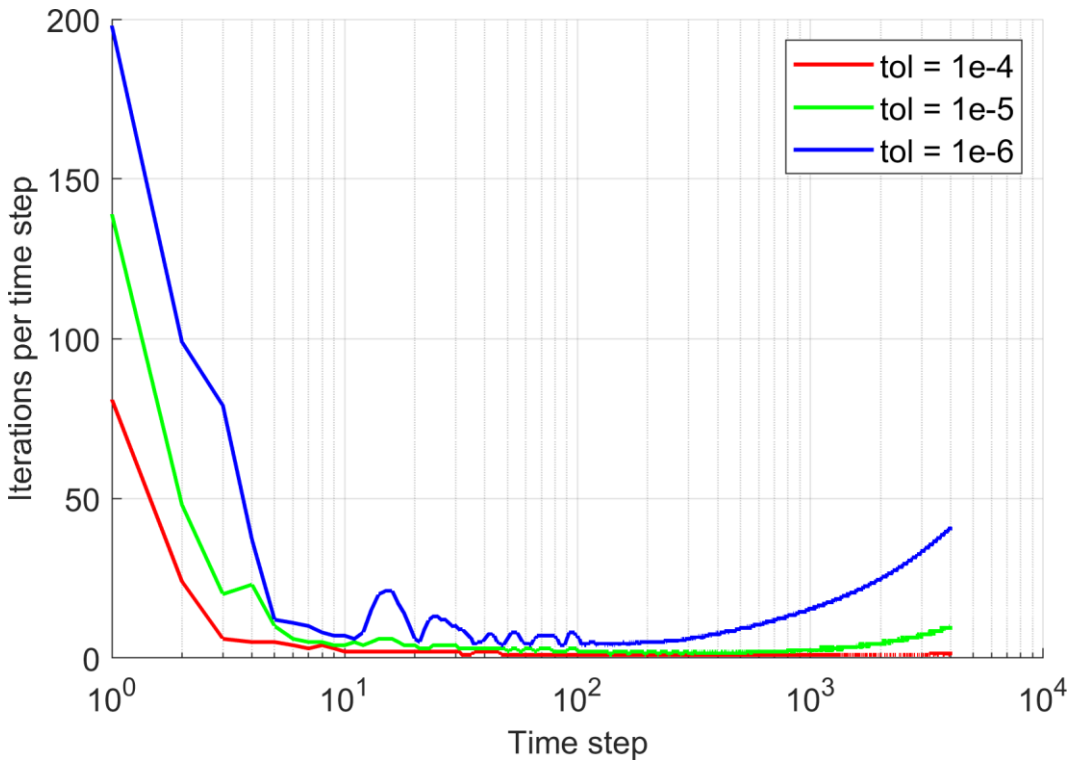

Figure 7: Number of iterations per step at different tolerance values for the shear case.

The average number of iterations is 1.07 when the tolerance is 10^{-4}, 5.05 when the tolerance is 10^{-5}, and 24.04 when the tolerance is 10^{-6}. A learge number of iterations

are required at the beginning because during these early time steps there is the largest pressure gradient. This makes the divergence of the provisional velocity relatively high in localised parts of the domain, which requires more iterations to converge. After 100 time steps the number of iterations is relatively stable for the tolerances 10^{-4}, although there are noticeable increases for the 10^{-5} and 10^{-6} cases, which could be due to the progressive development of the circulating regions. In any case, the iteration count is not 1 because the flow has not reached a steady state. From the comparison of the outputs between the iterative and direct solver for a tolerance of 10^{-4}, the solutions already compared well to the direct solver. This means that the errors of order of approximately 10^{-4} persisting in the later time steps require more iterations to reduce, despite having a limited impact on the solution.

2.2.3 Solver comparison

To compare the performance of the direct and iterative solvers, the total time for each solver was measured. In each case, the computation time was commenced prior to initialisation and stopped after 100 time steps (from Figure 7, 100 time steps was when the iteration numbers had stabilised, so the time steps beyond this were excluded given that both methods would perform these comparatively quickly). The time starting prior to initialisation was done to include the cost of inverting the coefficient matrix in the time for the direct solver, albeit spread across all time steps. This was specifically undertaken for grid resolutions $[m, n] = [(20, 20), (40, 40), (60, 60), (80, 80), (100, 100)]$. The time comparison is shown in Figure 8.

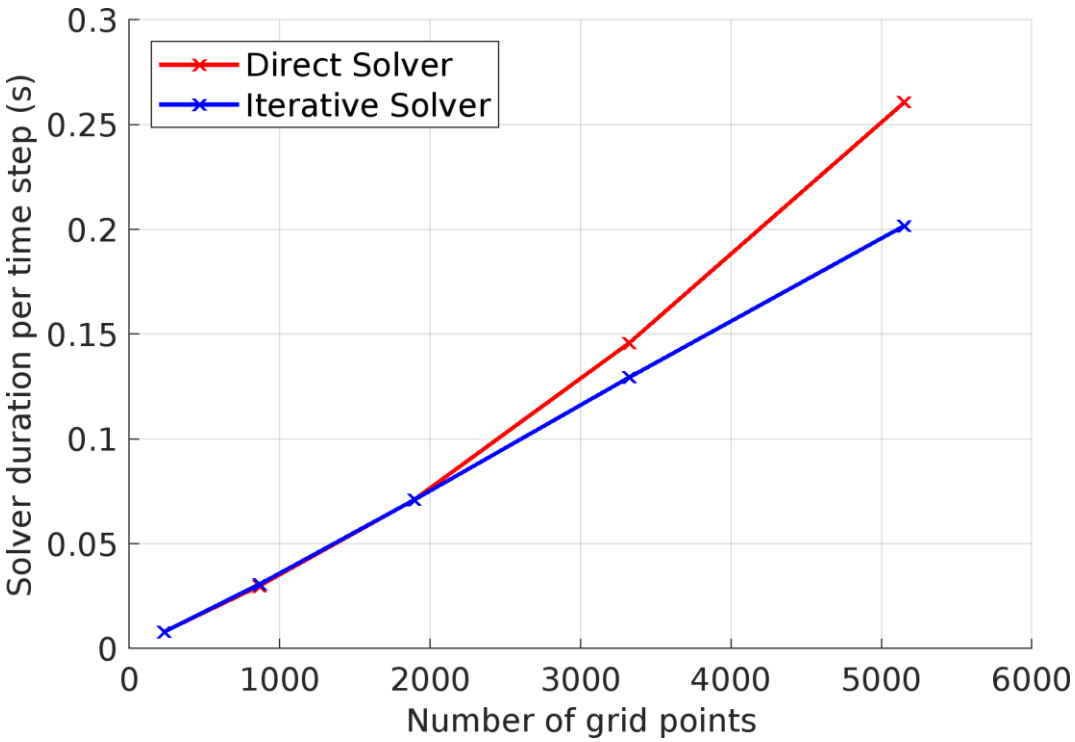

Figure 8: Comparison of the average computation time per time step between the direct and iterative solution methods for the shear case.

The time consumed per time step by the iterative solver is smaller than the direct solver for all resolutions, and the time has a nearly linear relation to the total number of grid points (demonstrating solver duration scales with n^2 where n is the number of points in one dimension). The time consumed by direct solver remains almost the same as the iterative solver at first, and then shows a larger value when the number of grid points over 2000. This is because the direct solver scales with n^3. The difference in solver performance for larger numbers of grid points is because it becomes very computationally expensive to invert the coefficient matrix, and even when this cost is spread over all time steps it is still relatively more expensive than the iterative solver.

3 Channel Flow

3.1 Problem description

The second flow case is 2D channel flow. The domain is defined on the range $x \in [0, 10]$ and $y \in [0, 2]$, where x represents the length of the channel and y the width (it is assumed there is no variation throughout the depth). Periodic boundary conditions were specified on $x = 0, x = 10$, and no-slip boundary conditions were specified on $y = 0, y = 2$. The velocity fields were initialised as given by (14) and (15)

$$u_0(x, y) = 0 \tag{14}$$

$$v_0(x, y) = \frac{1}{100} sin(\pi y) cos\left(\frac{\pi x}{10}\right) \tag{15}$$

Additionally, $p(x, y, t = 0) = 0$. The initialisation of $v_0(x, y)$ with small values ($v_0 < 0.01$) was done to examine the behaviour of the numerical solution rather than to represent a real phenomenon. This is shown in Figure 9.

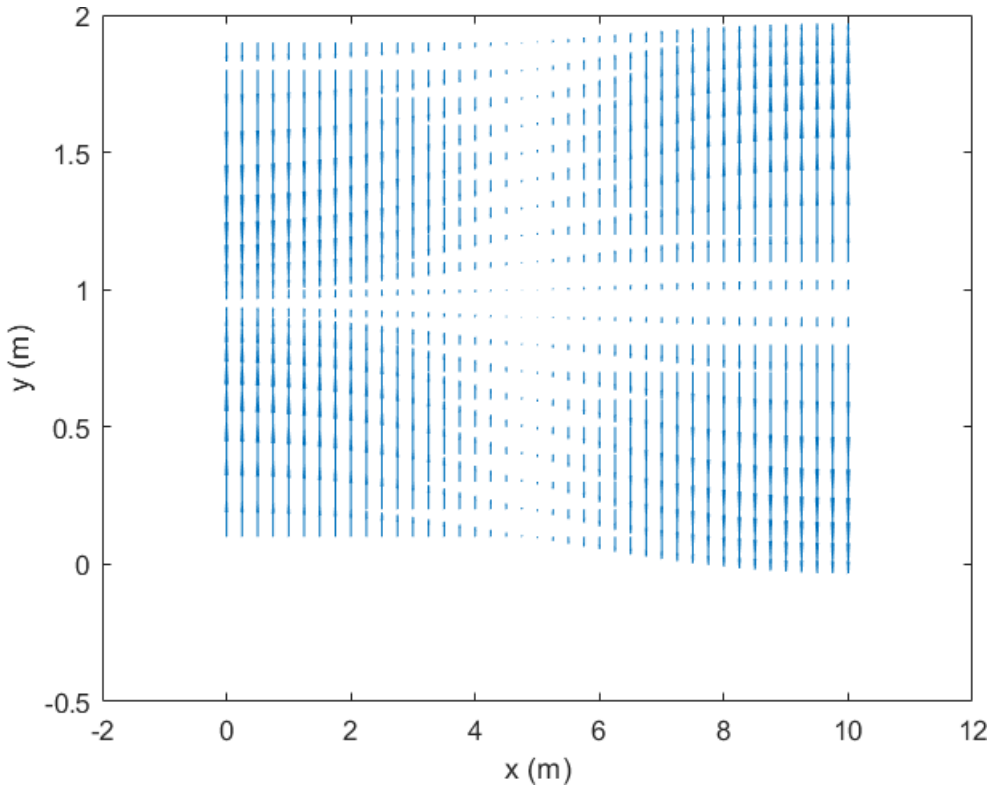

Figure 9: Initial velocity field in channel.

3.2 Numerical experiments

3.2.1 Direct Solver

The first numerical experiment was run using $\Delta x = 0.5m$ and $\Delta y = 0.2m$, with a 21 x 11 spatial grid. The kinematic viscosity and density were set as $v = 1m^2/s$ and $\rho = 1kg/m^3$ for all numerical experiments. The initial time step was chosen as $\Delta t = 0.0025s$.

The flow field is studied at multiple time steps. At each time step, the contours of the velocity magnitude and the velocity vectors are plotted. To better visualise the velocity profile, the $u(x = 0, y)$ is plotted in red. The results of the first experiment are shown in Figure 10.

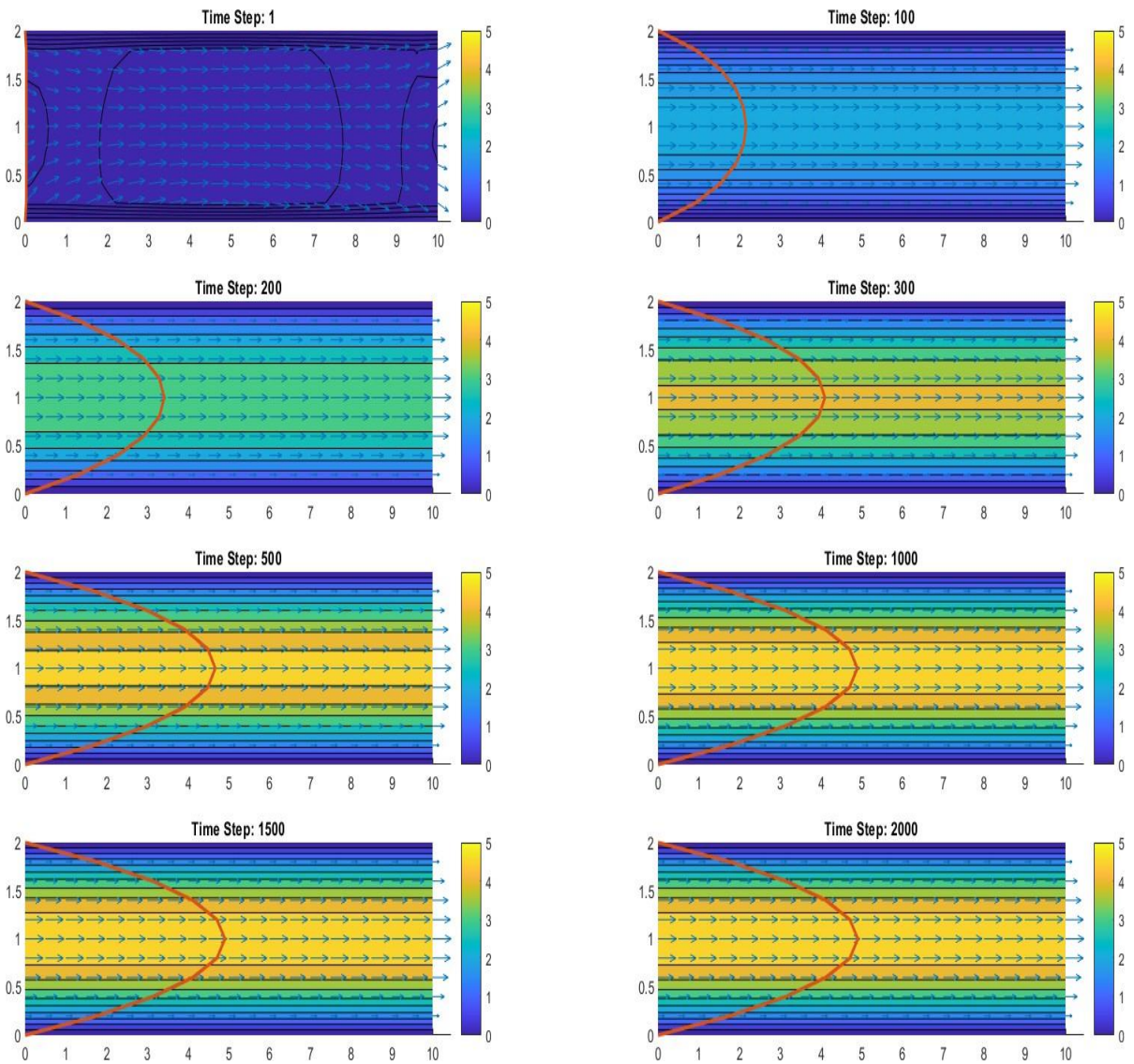

Figure 10: Flow field at different time steps for Δx = 0.5m, Δy = 0.2m, Δt = 0.0025s.

It was observed that v rapidly decays to negligible values, and u quickly grows. The gravity term causes high acceleration of the flow in the x-direction, although the acceleration of the flow decreases. A high velocity gradient, $\frac{\partial u}{\partial y}$, is established, causing high shear stress which counteracts the acceleration due to gravity.

To study whether the flow reached steady state, the u velocity was measured at the center of the channel. The criterion for steady state was a change in the channel's center-line velocity of less than 0.1% for a single time step. The simulation is continued for a fixed number of iterations but the time step of convergence is marked with a triangle in Figure 11. After 2.6375s, or 1055 time steps, the criterion for steady state has been met. Beyond this point, the flow is considered fully developed.

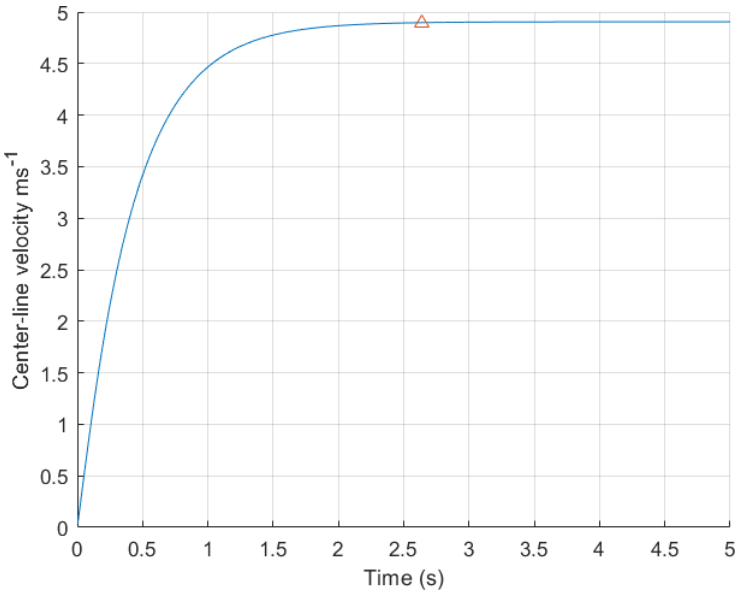

Figure 11: Center-line velocity in channel for the first numerical experiment.

The pressure field is shown in 12. A time step was chosen which best demonstrated the issue. Non-physical oscillations in the pressure field can be observed. These oscillations are a result of the discretisation as discussed in the introduction. The pressure at any given point is only coupled to the velocity at the neighbouring points, and vice versa, which means non-physical pressure oscillations with a wavelength of $2\Delta x$ can exist within the solution. The pattern of oscillations is consistent with the theory, especially when examining the difference in pressure between the neighbouring points in the zoomed-in subdomain in Figure 12.

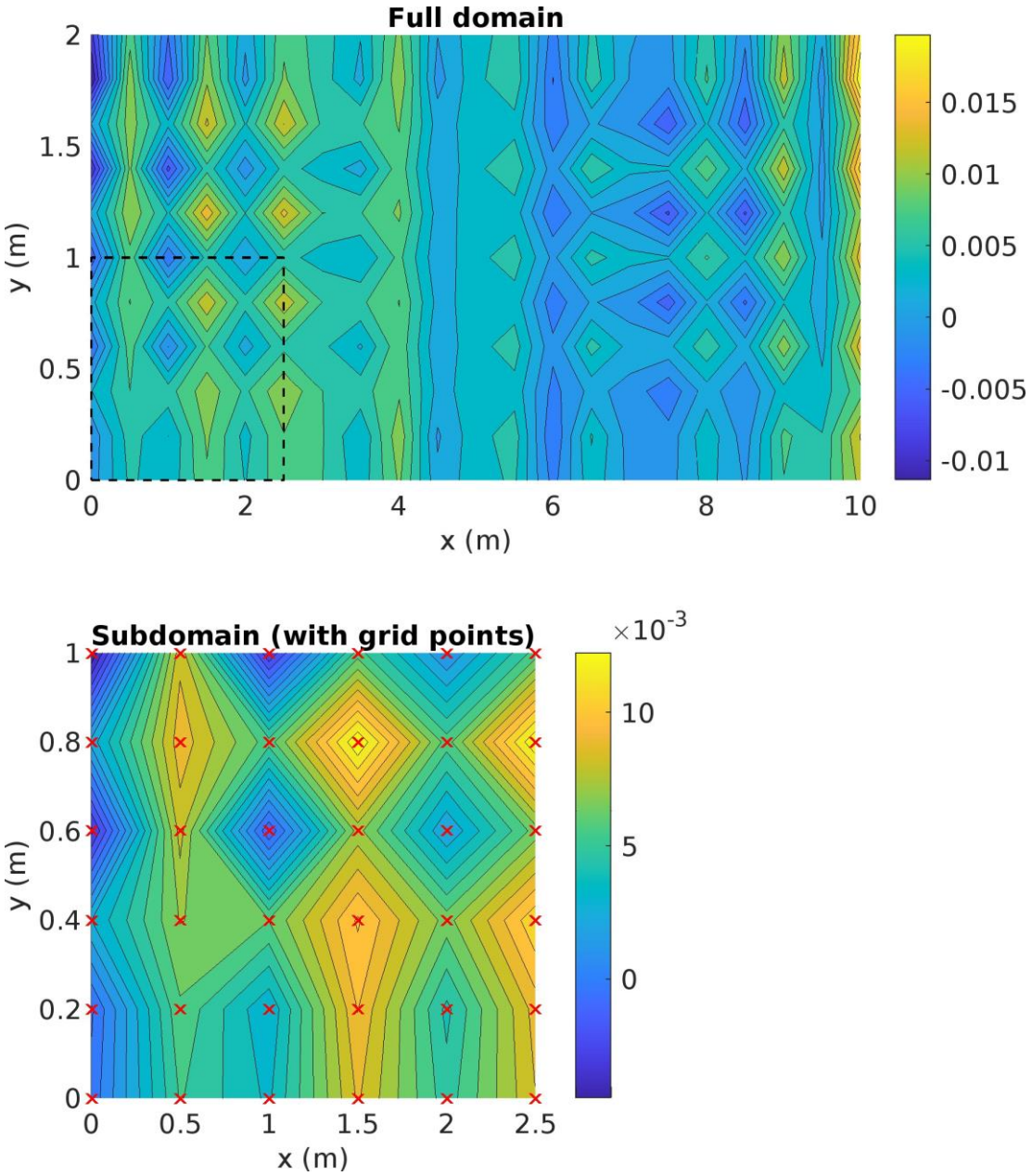

Figure 12: Contours of the pressure field in the channel with the 21 x 11 grid at time step 20

The next experiment used $\Delta x = 0.25$ m and $\Delta y = 0.10$ m, corresponding to a 41 x 21 grid. The time step remains same at 0.0025 s. The resulting velocity profile at different times matched the previous experiment (plots of the results are excluded for brevity). It was found that the steady state criterion was reached slightly earlier at at 2.625 s (1050 time steps). The higher resolution grid has a lower discretisation error, which accounts for the minor difference, although the magnitude of the difference is only small and indicates both solutions are good approximations. However, the pressure oscillations were still present. Furthermore, the pattern confirmed that the oscillations are dependent on

the grid, which means the phenomenon is non-physical. The pressure oscillations are shown in Figure 13.

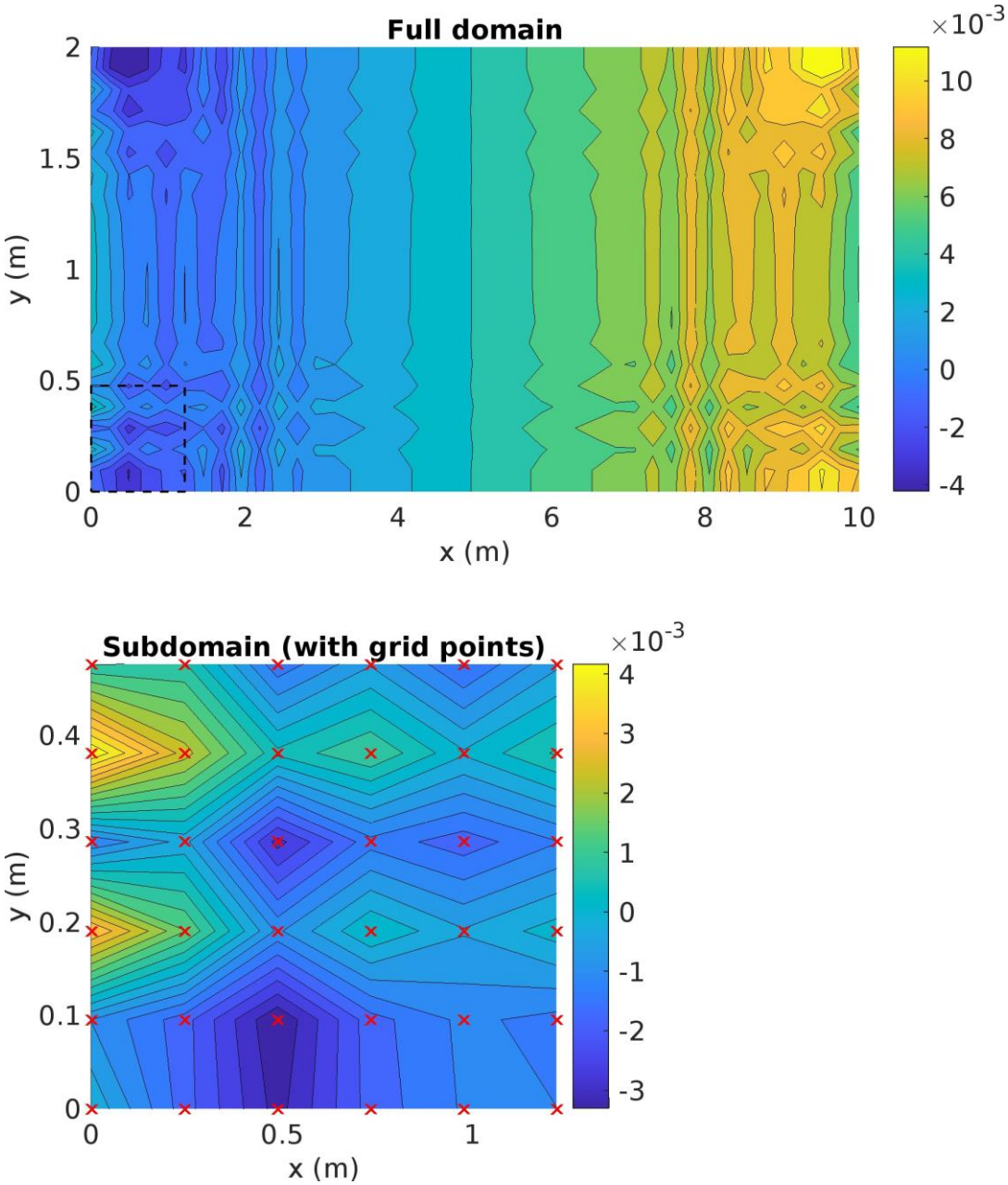

Figure 13: Contours of the pressure field in the channel with the 41 x 21 grid at time step 20

For the next experiment, the grid was refined further to $\Delta x = 0.25$ m and $\Delta y = 0.10$ m (a 61 x 31 grid). The solution values became unbounded after 100 time steps due to the CFL number being too high, causing instability. In this case, the Courant-Friedrichs-Lewy (CFL) number for a 2D grid is given by (16)

$$CFL = \frac{u_{\max}\Delta t}{\Delta x} + \frac{v_{\max}\Delta t}{\Delta y} \tag{16}$$

where $u_{\max}$ is the maximum velocity in the x direction and $v_{\max}$ is the maximum velocity in the y direction. In this case, since $v_{\max}$ is 0, so Δy does not affect the CFL number. To verify this, the next numerical experiment is conducted on a grid of 41 x 31,and no stability issues were encountered.

For the next experiment, the time step was reduced to 0.0015s for the refined grid (61 x 31) and the same qualitative solution was obtained. A steady-state centreline velocity occurs at 2.4135 s (1609 time steps). The finer temporal resolution has further reduced the discretisation error, which is why the result is slightly different to the previous experiments.

3.2.2 Iterative Solver

The solution to the channel problem was approximated using the same iterative solver, the Gauss-Seidel method, for the pressure correction step. The maximum number of iterations per time step was set to 1000 and the tolerance was set to 10^{-4}. The same steady-state criterion for maximum velocity in successive iterations was applied. To validate the Gauss-Seidel method, the results obtained were compared with those from the direct solver. The parameters were kept the same as the first numerical experiment described in the preceding section. The difference in velocities at different time steps for the reference case are plotted as a contour and are shown in Figure 14.

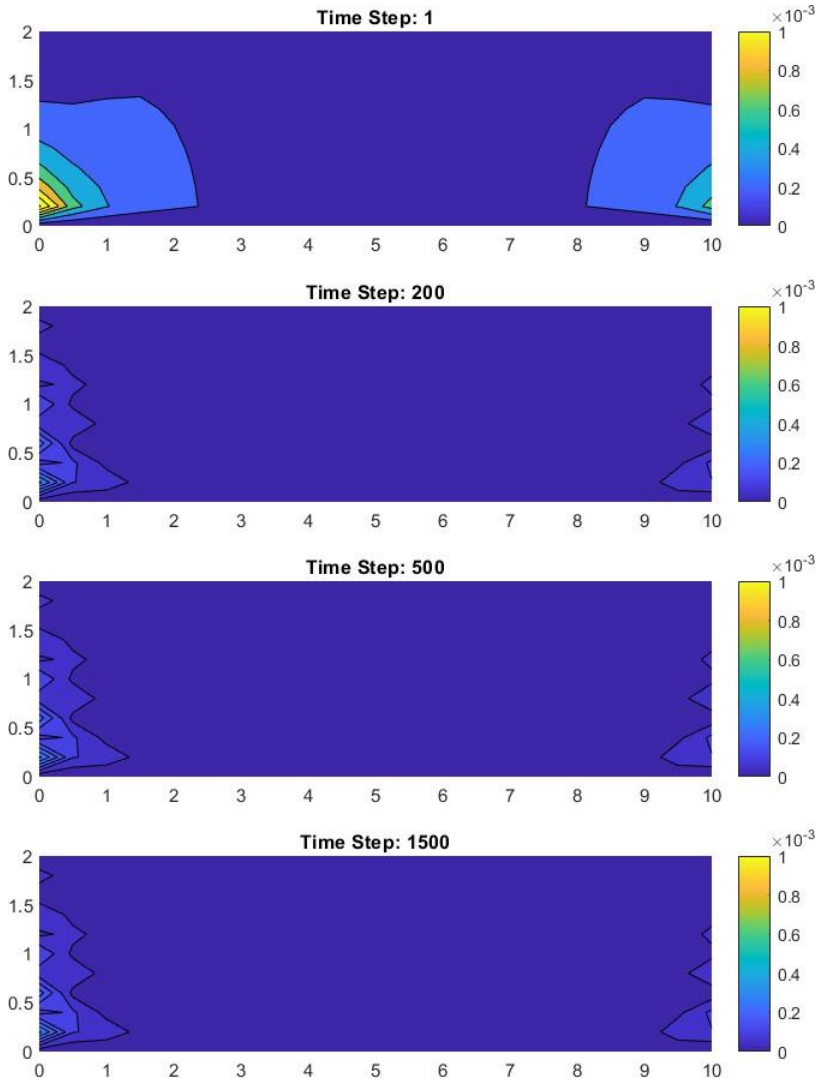

Figure 14: Absolute error in velocity with Gauss-Seidel method compared to the direct solver.

The largest errors are visible at the earliest time step, which can be explained by this being the time where the dynamics of the system are causing the larges changes in velocity in a single time step. At this step, as will be discussed, the maximum number of iterations is reached, and a residual error is present. However, the magnitude of the error is low. Given that the maximum error at any time step is less than 10^{-4}, the Gauss-Seidel solver is considered capable of achieving an appropriate approximation to the solution when compared to the direct solver.

The parameter influences on the number of iterations were investigated, with the similar influences present for the channel. No discussion is presented for brevity.

The number of iterations per time step for channel flow is shown in Figure 15.

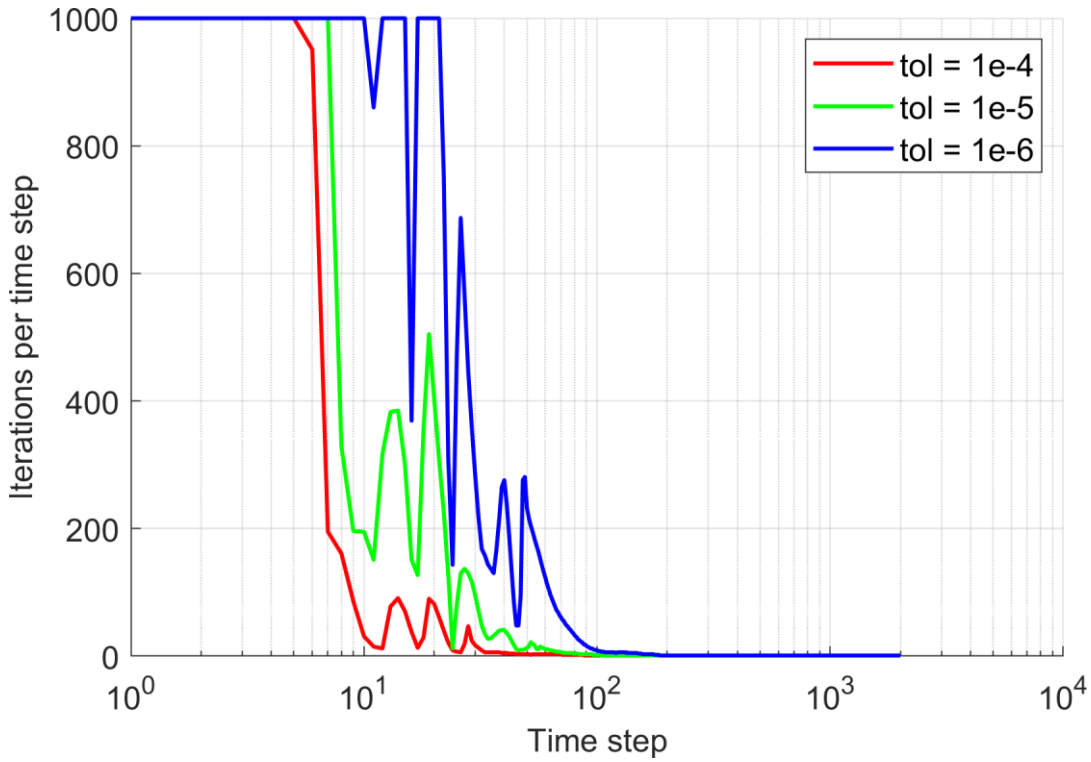

Figure 15: Number of iterations per step at different tolerance values for the channel case.

The results are plotted for three different tolerance values. The average number of iterations is 4.65 when the tolerance is 10^{-4}, 7.48 when the tolerance is 10^{-5}, 16.62 when the tolerance is 10^{-6}. These averages are all consistently higher than the averages for the shear case, indicating the greater computational demand of the problem on the iterative solver. For all scenarios, a high number of iterations are required at the earliest time steps, and even for the largest error tolerance the maximum number of iterations is reached. After 100 time steps, the number of iteration is relatively small for 10^{-4}, 10^{-5}, 10^{-6} cases. This is related to the flow fields developing rapidly over the first hundred time steps but then reaching a steady state, where the guess of the pressure correction (of zero everywhere) is actually very close to required adjustment.

3.2.3 Solver Comparison

In the same manner as for channel flow, the time consumed per time step on average by each of the solver methods was plotted in Figure 16.

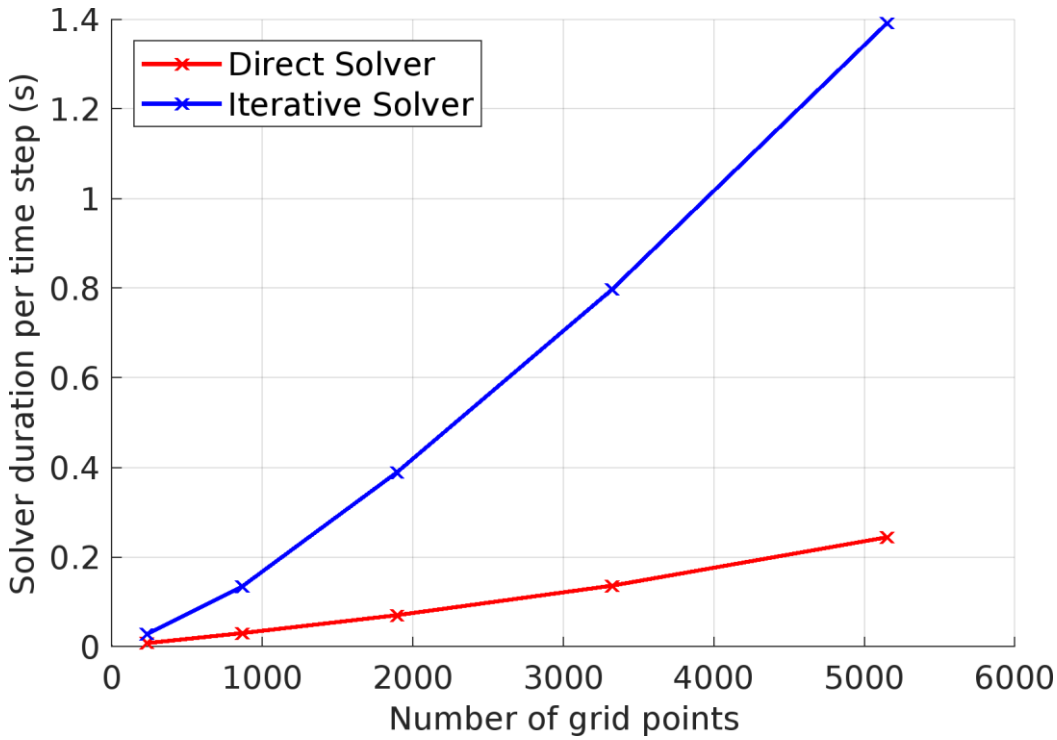

Figure 16: Comparison of the average computation time per time step between the direct and iterative solution methods for the channel case.

In contrast to the shear flow case, the time consumed per time step by the direct solver is smaller than the time cost by iterative solver. This occurs because the iterative solver, in each of these scenarios, has many time steps where the maximum number of iterations are required. Despite the computationally expensive inversion of the coefficient matrix for fine grid resolutions, the computation of the matrix only occurs once at the beginning of the solution procedure, meaning subsequent time steps are relatively cheap.

Comparing, the channel and shear flow case, the direct solver consumes a similar amount of time in each case, and it is the iterative solver that is significantly different between the two cases. This is because the time taken for the inversion the coefficient matrix (same size in each case) and the computation of each time step is independent of the specifics of the flow scenario, bearing in mind that these two scenarios are discretised in the same way. However, the iterative solver is impacted by the flow scenario because the required size of the pressure correction compared to the initial guess is critical for the number of iterations required. In the channel case, there is a particularly fast change in the flow velocity over the early time steps, which causes the high iteration count.

4 Conclusion

This report examined two flow cases using a direct and an iterative solver for the pressure correction step. Both solvers gave matching solutions in each of the numerical experiments, and the solutions obtained with varying parameters agreed with the expected results considering the physics governing the flow cases. The channel case exhibited checkerboard pressure oscillations, whereas a particular flow case for the shear flow exhibited wiggles in the velocity field, which are the result of the use of a collocated grid. These issues could be resolved with the use of a staggered grid or Rhie-Chow interpolation.

The performance in terms of computational time consumption per time step was compared for the direct and iterative solvers. It was found that the each solver had similar performance for the shear case. However, for the channel case iterative solver required more time per time step due to the high number of iterations required at early time steps. This was attributed to the dependence of the iterative solver on the specifics of the case for the demand on iterations per time step.